AA528
Spacecraft Dynamics & Control
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Mehran Mesbahi, Aeronautics & Astronautics, University of Washington

Introductory Remarks

I wrote these notes in different pieces- the last recent piece came during the Winter of 2025, right before I was about teach AA528 at the University of Washington.[1] The course is a graduate level course on spacecraft dynamics and control. The reason that I thought these notes would be useful is that although there are a handful of excellent textbooks on the topic, some of the students in class have historically struggled to make a coherent view of the field after only taking one quarter of spacecraft dynamics and control. This has been particularly the case for students who come to the field from outside of control and dynamics. Given the current rapid evolution of space systems (reusable launch vehicles, lunar habitats, planetary science, satellite constellations, to name a few) and a more heterogeneous group of engineers who are and will be working in the space engineering domain, this issue will only become more pressing in the coming decade. Common complaints are notation (even I have personal complaints about this), breadth of the material covered, inconsistent mathematics, background that is typically assumed, as by its nature, topics covered in the course touch upon a number of fields that might be unreasonable to assume from the students. My attempt has been to gather and provide a concise, mathematically clear way to reason about spacecraft dynamics and control. These notes will also set the foundation for other relevant topics that I had to cut from these lectures but will eventually appear on my website.

The topics that I will discuss in these lectures include the following–depending on how the actual lectures go– I will shorten or expand on some of these topics in real-time.

1. **Week 1**: Coordinates and Orientations: coordinate frames, rotation matrices, groups; quaternions and some algebra
2. **Week 2**: Taking derivatives, kinematics for translation and rotation; dynamics for translation and rotation
3. **Week 3**: stability of torque free motion; two-body problem
4. **Week 4**: landing, quaternion feedback, gravity gradient stabilization, introduction to rendezvous and docking
5. **Week 5**: Rendezvous and docking; two-body Part II/perturbation theory

[1] The date of the last update in included in the center footer.

6. **Week 6**: Attitude determination and sensing
7. **Week 7**: Attitude control and actuation
8. **Week 8**: Restricted three body problem and Halo orbits
9. **Week 9**: Constellations, GPS, space networks
10. **Week 10**: Project presentations

What is Spacecraft Dynamics and Control?

So I have listed the topics above but did not tell you why I have chosen these topics as opposed to many other that I could. It all boils down to problems, that in my view, form the core of spacecraft dynamics and control. For me, whose mind has been shaped (molded) by systems and control, the topics often involve obtaining expressions of the form,

$$\dot{\boldsymbol{x}}(t) = f(\boldsymbol{x}(t), \boldsymbol{u}(t), \boldsymbol{w}(t))$$

where $\boldsymbol{x}$ is "some" state associated with the spacecraft and $\boldsymbol{u}$ is some sort of control; $\boldsymbol{w}$ on the other hand, designates external influences on this dynamics that are not directly controlled. In "traditional" spacecraft dynamics and control, we go through various coordinate transformations to understand what f should be- in fact, even to be able to describe a spacecraft-related phenomena, we often need to be clear about coordinate frames. So, coordinate frames and how to parameterize them becomes one of the first things that we have to settle.

At some point, particular when we think about control, we will also need to make measurements about this system; these measurement assume the general form,

$$\boldsymbol{y}(t) = h(\boldsymbol{x}(t), \boldsymbol{u}(t), \boldsymbol{v}(t))$$

where $\boldsymbol{v}(t)$ are noise terms on these measurements. Attitude determination for example involves the process of going from the corresponding $\boldsymbol{y}$'s to estimate the state of interest (orientation of the spacecraft).

For those of you have already taken control courses, it is no coincidence that the above two equations are often the starting point for control/system design–we will not only derive these equations for spacecraft settings (which I think is not only fun but also creative), in order to do things like stability analysis, but also design trajectories or feedback control or estimation algorithms for certain spacecraft-related classical problems.

Okay- of course this is all high level ... so I will sprinkle the lecture with spacecraft related problems, describing missions and scenarios of interest, throughout the course to motivate the mathematics and the design.

Expectations and assessment

The background expectation for you is some basic knowledge of linear algebra, ordinary differential equations, and mechanics- I will review some of these quickly and even include a bit of twist on these

topics from time to time. I also expect that you can code things up in some high level language like Python, Julia, Matlab for most of the homework assignments; this is really important. You should be a able to write code to simulate dynamical systems, solve linear algebra problems, and before the course is done, code up optimization-type routines- coding is good for you.

We will have weekly homework assignments; the reason for this is that math and engineering are really not meant to be "watched"- well, that is really true for everything- you can't become a soccer player by watching MLS or premier league. Easy homework problems boost confidence, hard ones make you develop more intuition (and build character)- not that we have very difficult problems assigned in this course but "difficulty" is relative- a reflection of previous experiences and background, knowledge, and natural affinity for a concept- the latter one is sometimes referred to as aptitude, but it is hard to quantify. I think hard work goes a long way regardless.

So, back to homework- you will see the problems at the end of every lecture (two lectures per week) but the assigned problems will be due every Friday starting the second week of the course (so for example, the problems at the end of Lectures 1 and 2 will be due the second Friday into the course)- I will include a catalyst link for submitting homework (due at 11:59 pm on every Friday except the first one into the quarter); we have office hours on Thursdays from 4-6pm to discuss homework; we can call these homework clinic- if you are wounded by working on the homework you can come to these sessions to be healed. But try solving these problems before you show up to these clinics- it is important that you attempt these problems before you seek a suggestion- pedagogically this is the only way you can learn.

We will also have a project component for the course- more on this later. The grading for the course will be 65% based on homework and 35% based on the project.

On references

In addition to these notes, I have found having a book handy as a reference is useful for students as notes/lectures can't cover everything- as such, we have a textbook adopted for class and I will also use to assign homework from. The textbook that we will refer to by "MC2014" is,

> F. Landis Markley and John L. Crassidis, *Fundamentals of Spacecraft Attitude Determination and Control,* Springer, 2014.

Note that the notation that I use sometimes is different from those used in this book; I will only change the notation on you if I have a strong preference for one over the other: in my view, notation be insightful, but also elegance and clean, and the later two features, I do understand, are personal. Nevertheless, for the purpose of this class, I have adopted the notation in the book, when the probability of confusion is higher than the perceived benefit- for example, I like to use $\boldsymbol{\tau}$ for external torque on the spacecraft; the book uses $\mathbf{L}$; my compromise has been $\boldsymbol{L}$. But there are also notation that that I have not compromised on- for example, direction cosine matrices for coordinate transformation are denoted like $A_{FF'}$ (where F and F' are two distinct frames); I use $R^{\mathbf{ba}}$. I find having three capital letters aesthetically non-pleasing; coordinate frames for me should be lower case letters if they are appearing next to an upper case as super or sub scripts. I list a few more below:

MM notation	MC2004 notation	description
$\mathbf{i}, \mathbf{a}, \mathbf{b}, \cdots$	$F, F', R, \cdots$	coordinate frames
$R^{\mathbf{ba}}$	$A_{FF'}$	direction cosine matrix

Here are other references that I have consulted to put these notes together- they are all good and recommended for you to enjoy:

- Analytical Mechanics of Space Systems (Fourth Edition) by Hanspeter Schaub and John L. Junkins.
- Orbital Mechanics for Engineering Students (Fourth Edition) by Howard D. Curtis.
- Fundamentals of Astrodynamics by Roger R. Bate, Donald D. Mueller, and Jerry E. White; this is a classic, often referred to as *BMW*.
- 3Blue1Brown
- Alfonso Gonzalez - Astrodynamics & SE Podcast

I will add more references to the above list over time as needed. But looking up too many references at least in the beginning of the course can become confusing. Enjoy them over time- these notes should suffice for the most part but I will include some remarks on relevant material on the notes as well.